Ex No:02

30.07.26

Study about different types of network cables.

Aim:-

To study about different types of network cables.

Different types of Cables:-

1. Unshielded Twisted Pair (UTP) cable
2. Shielded Twisted Pair (STP) cable
3. Coaxial cable
4. fibre Optic cable.

Cable Type: UTP

Category	Max. Data Transmission
Category 3	10 bps
Category 5	Up to 100 Mbps
Category 5e	1 Gbps

Advantages:-

- Cheaper in cost
- Easy to install as they have a
- Small overall diameter.

Disadvantages :-

- More prone to Electromagnetic interference and noise.

Application Use :-

- 10-Base-T Ethernet
- fast Ethernet, Gigabit Ethernet.

Cable Type :- STP

Category	Max. Data Transmission
Category 6, 6a	10 Gbps
Category 7	10 Gbps

Advantages :-

- Shielded
- faster than UTP
- less susceptible to noise and interference.

Disadvantages :-

- Expensive
- Creates Installation effort.

Application :-

- Category 6/6a ⇒ data centres.
- Category 7 ⇒ Gigabit ethernet (100m)

Cable Type :- Coaxial Cable

Category : RG-6, RG-59, RG-11

Max data Transmission : 10 - 100 Mbps

Advantages :-

- High bandwidth
- Immune to interference
- Low loss bandwidth
- Versatile

Disadvantages :

- Limited distance, cost
- Size is bulky

Cable Type: fibre Optics Cable

Category	Max. data Transmission
Single Mode Multi Mode	100 Gbps

Advantages :-

- High speed & bandwidth
- high security

Disadvantages :-

- Expensive
- Requires skilled installers.

Application/Use :-

Maximum distance of fibre optics cable is around 100 metres.

Result :-

Thus the study about the different types of cables are executed successfully.